

Functional and Spectral Foundations of the Scalar Laplacian on the Menger Sponge

A Mathematical Basis for the TCA-F5D Program

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Paper A

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Geometry determines the energy; the energy determines the operator; the operator determines the spectrum. The spectrum is not constructed to reproduce a predetermined value.

Abstract

This article establishes a functional and spectral framework for the canonical scalar Laplacian on the standard Menger sponge and examines its use as an internal sector of a free scalar field associated with the TCA-F5D program. The construction starts from the self-similar metric measure space (M, ρ, μ) and the canonical class of regular, symmetric, strongly local, and irreducible Dirichlet forms, unique up to a positive multiplicative constant. The global energy scale is fixed by an explicit normalization convention based on the effective resistance between two opposite exterior faces.

The normalized form determines two nonnegative self-adjoint realizations: the conservative realization L_M^N and the exterior Dirichlet realization L_M^D . Their semigroups are trace class at every positive time, their resolvents are compact, and their spectra are discrete. Spectral positivity is expressed through the positive scalar measures induced by their spectral resolutions.

The short-time heat trace admits an expansion governed by the walk dimension d_w , the spectral dimension $d_s = 2d_H/d_w$, and coefficients periodic in logarithmic time, with period $\ln \tau_M$, where $\tau_M = 3^{d_w}$. Mellin transformation yields meromorphic continuations of the spectral zeta functions and locates candidate pole families; nonreal poles are effective only when the corresponding Fourier coefficients do not vanish.

The internal operator is finally incorporated into a free scalar field on $\mathbb{R}^{1,3} \times M$ in an operator-theoretic sense. The modal decomposition produces masses

$$M_{n,\#}^2 = m_0^2 + \Lambda_M^2 \lambda_n^\#.$$

No numerical values are assigned to d_w , d_s , or the spectrum; the modes are not identified with observed particles; and no universal physical constant is derived from $\zeta'(0)$. The guiding methodological principle is that the operator should determine the spectrum, rather than being adjusted to reproduce predetermined phenomenological values.

Keywords. Menger sponge; Dirichlet form; fractal Laplacian; spectral analysis; heat kernel; spectral dimension; spectral zeta function; log-periodicity; effective resistance; free scalar field; TCA-F5D.

Scope Statement

This work does not constitute a general validation of TCA-F5D. Its scope is restricted to the functional and spectral analysis of a model whose internal sector is represented by the standard Menger sponge. Mathematical results adopted from the literature, consequences derived within the model, and physical hypotheses that remain open are explicitly distinguished.

The numerical computation of the spectrum, the exact determination of r_M , d_w , and d_s , the identification of modes with particles, fermionic and gauge sectors, interactions, renormalization, dynamical gravity, consciousness, ultraviolet finiteness, and scattering unitarity lie outside the scope of this article.

Specific Contributions of the Manuscript

The existence and uniqueness-up-to-scale theorems, heat-kernel estimates, and heat-trace expansion are adopted from the cited literature and are not presented as original results. The specific contributions of this manuscript are organizational, specialized, and deductive:

1. to establish a coherent architecture for the standard Menger sponge, distinguishing the ambient boundary from the exterior boundary used in the Dirichlet realization;
2. to introduce, as an explicit and reproducible convention, the normalization $R_{01}(\mathcal{E}_M) = 1$ between two opposite exterior faces;
3. to derive explicitly the meromorphic continuation of the reduced Neumann zeta function from the heat-trace expansion and the Mellin transform;
4. to construct the product operator of the free scalar field and clarify the status of its modal thresholds, quadratic positivity, and spectral dimension per unit volume;
5. to formulate convergence and falsifiability criteria that separate the mathematical foundation from any future phenomenological identification.

Academic Continuity and Provenance

This manuscript is a critical reconstruction of the TCA-F5D program previously archived on Zenodo as a main article accompanied by three *Companion* documents [14]. The historical deposit is available at [doi:10.5281/zenodo.20652631](https://doi.org/10.5281/zenodo.20652631). Its role in the present work is to establish the documentary continuity of the program, not to serve as evidentiary support for the mathematical theorems employed here.

Claims from the earlier corpus are not imported as premises. Paper A retains only statements that have been reconstructed from explicit definitions, derived within the manuscript, or supported by primary mathematical sources. Whenever an earlier formulation is incompatible with the scope of the present work, Paper A supersedes it for the functional and spectral sector studied here.

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1. Introduction, Scope, and Fundamental Question

1.1. Context and Objective

Spectral analysis on fractal spaces replaces part of the classical differential apparatus with a formulation based on metric measure spaces, Dirichlet forms, and self-adjoint operators. A self-similar fractal generally lacks an ordinary differentiable structure on which one could directly introduce a smooth metric tensor, a Levi-Civita connection, or a Laplace–Beltrami operator. It may nevertheless support a regular Dirichlet form that determines a diffusion, a semigroup, and a Laplacian in the functional-analytic sense [3, Theorem 1.3.1][2, Theorem 1.2 and Corollary 1.3].

This article studies that architecture for the standard Menger sponge $M \subset [0, 1]^3$, equipped with the restricted Euclidean metric ρ and the uniform self-similar measure μ . The work continues and revises the TCA-F5D corpus deposited on Zenodo [14]; that deposit is cited as an academic antecedent and development record, whereas the mathematical justification of the present results relies exclusively on the primary sources and internal derivations indicated in each section. The construction follows the chain

$$(M, \rho, \mu) \longrightarrow (\mathcal{E}_M, \mathcal{F}) \longrightarrow L_M^\sharp \longrightarrow e^{-tL_M^\sharp} \longrightarrow K_M^\sharp(t) \longrightarrow \zeta_M^\sharp(s), \quad (1.1)$$

where $\sharp \in \{N, D\}$ distinguishes the conservative realization from the exterior Dirichlet realization. The objective is to determine the functional, spectral, and thermal structure generated by this chain and the conditions under which it may be incorporated into a free scalar field.

1.2. Geometric Hypothesis and Methodological Principle

The Menger sponge provides an explicit three-dimensional self-similar compact set defined by twenty similarities of ratio $1/3$. It also belongs to the class of generalized Sierpiński carpets for which results are available on Dirichlet forms, diffusion, walk dimension, heat kernels, and heat traces [5, Definition 2.2][2, Theorem 1.2][6, Theorems 4.7–4.10]. Its faces, edges, and vertices further make it possible to distinguish the strata entering the subleading terms of the heat trace.

The choice of M is a concrete hypothesis of the model, not a physical conclusion. This work does not prove that the sponge is a unique internal geometry, that it is dynamically selected, or that it is physically realized in nature. The conditional statement used throughout Paper A is:

If the internal sector is modeled by M , its scalar operator must be constructed from the corresponding canonical Dirichlet form.

The methodology is summarized by

$$\text{geometry} \longrightarrow \text{energy} \longrightarrow \text{operator} \longrightarrow \text{spectrum}. \quad (1.2)$$

Accordingly, the spectrum is not introduced to reproduce a constant, a dimension, or a mass ratio selected in advance. The mathematical normalization of the energy will be fixed through effective resistance; the physical scale will be introduced separately through $\Lambda_M > 0$:

$$\text{mathematical normalization} \neq \text{physical determination of scale}. \quad (1.3)$$

1.3. Fundamental Question and Contributions

The question organizing the article is:

What rigorous mathematical architecture can be established from the canonical scalar Laplacian on the Menger sponge for the TCA-F5D program, and under what conditions can it be used as an internal sector of a free scalar field on $\mathbb{R}^{1,3} \times M$, without imposing its spectrum or a phenomenological scale a priori?

To answer this question, the article:

1. defines the geometry, measure, and a distinguished exterior boundary;
2. adopts the canonical class of Dirichlet forms and fixes its global scale by $R_{01} = 1$;
3. constructs the self-adjoint realizations L_M^N and L_M^D ;
4. establishes compactness, spectral discreteness, and positivity of the diagonal spectral measures;
5. organizes the heat-trace expansion, spectral dimension, and meromorphic continuation of the zeta functions;
6. incorporates the internal operator into a free scalar field and derives

$$M_{n,\sharp}^2 = m_0^2 + \Lambda_M^2 \lambda_n^\sharp. \quad (1.4)$$

The contribution is not to assign values to the eigenvalues, but to establish the mathematical framework within which they must be computed.

1.4. Epistemic Status and Exclusions

Three levels are distinguished: adopted external results, consequences derived in this work, and open physical hypotheses. The first level includes existence and uniqueness up to scale of the canonical form, energy self-similarity, $d_w > 2$, sub-Gaussian estimates, and the heat-trace expansion. The second includes the boundary realizations, compactness of the resolvents, positive spectral measures, the reduced Neumann zeta function, and the modal tower. The physical interpretation of M , the determination of Λ_M , and any correspondence with particles remain open.

The analysis is restricted to the scalar Laplacian and a free quadratic theory. It does not construct Dirac operators, fermionic fields, chiral structures, gauge sectors, interactions, dynamical gravity, renormalization, or an S -matrix. Nor does it postulate $r_M = 3/5$, $d_s \simeq 1.72$, or a thermal period $\ln 3$; it does not derive a universal physical constant from $\zeta'(0)$; and it does not claim ultraviolet finiteness or interacting unitarity.

1.5. Organization and Closing Statement

Sections 2 and 3 establish the geometry, measure, canonical form, and its normalization. Sections 4 and 5 construct the operators and their spectral structure. Sections 6 and 7 develop the heat trace and the zeta functions. Section 8 formulates the free scalar field. Section 9 discusses physical scope and falsifiability criteria, and Section 10 presents the conclusions and the subsequent program.

Before physical meaning is assigned to a fractal spectrum, the geometry and measure must be defined, the energy must be constructed and normalized, the operator must be identified, and the convergence of the procedures used to compute its spectrum must be established. Paper A provides the rigorous architecture corresponding to the geometry, measure, energy form, and operator. The construction of convergent approximants and the numerical discovery of the spectrum will be the subject of Paper B.

2. Self-Similar Geometry, Measure, and Boundary Notions

2.1. Self-Similar Construction

Let $Q_0 = [0, 1]^3$ and

$$\mathcal{S}_M = \left\{ \mathbf{i} = (i_1, i_2, i_3) \in \{0, 1, 2\}^3 : \sum_{j=1}^3 \mathbf{1}_{\{1\}}(i_j) \leq 1 \right\}. \quad (2.1)$$

The set $\mathcal{S}_M$ contains the twenty subcells that remain after removing the central cube and the six cubes centered on the faces. For each $\mathbf{i} \in \mathcal{S}_M$, define

$$F_{\mathbf{i}}(x) = \frac{x + \mathbf{i}}{3}. \quad (2.2)$$

By the attractor theorem for an iterated system of contractions [1, Section 3.1], the standard Menger sponge is the unique nonempty compact set satisfying

$$M = \bigcup_{\mathbf{i} \in \mathcal{S}_M} F_{\mathbf{i}}(M). \quad (2.3)$$

For a word $\omega = (\mathbf{i}_1, \dots, \mathbf{i}_n) \in \mathcal{S}_M^n$, define

$$F_{\omega} = F_{\mathbf{i}_1} \circ \dots \circ F_{\mathbf{i}_n}, \quad M_{\omega} = F_{\omega}(M), \quad Q_{\omega} = F_{\omega}(Q_0). \quad (2.4)$$

The geometric approximants are

$$Q_n = \bigcup_{\omega \in \mathcal{S}_M^n} Q_{\omega}, \quad M = \bigcap_{n \geq 0} Q_n. \quad (2.5)$$

2.2. Metric and Hausdorff Dimension

The space M is equipped with the restricted Euclidean metric $\rho(x, y) = |x - y|$. Since the twenty similarities have ratio $1/3$ and the system satisfies the open set condition, Hutchinson's self-similar dimension formula [1, Sections 5.2–5.3] gives

$$d_H = \frac{\ln 20}{\ln 3} \approx 2.726833. \quad (2.6)$$

Topological dimension and Hausdorff dimension play distinct roles; the latter describes measure scaling and does not by itself determine the diffusion.

2.3. Self-Similar Measure

The natural measure is the unique uniform self-similar Borel probability measure [1, Section 4.4] satisfying

$$\mu = \frac{1}{20} \sum_{\mathbf{i} \in \mathcal{S}_M} (F_{\mathbf{i}})_{\#} \mu, \quad (2.7)$$

so that

$$\mu(M_{\omega}) = 20^{-n}, \quad |\omega| = n. \quad (2.8)$$

This uniform self-similar distribution will be the reference measure in $L^2(M, \mu)$.

2.4. Empty Interior and Ambient Boundary

The three-dimensional volume of Q_n is

$$\text{Vol}_3(Q_n) = \left(\frac{20}{27}\right)^n \longrightarrow 0. \quad (2.9)$$

Therefore, M has empty Euclidean interior. Since it is closed,

$$\partial_{\mathbb{R}^3} M = M. \quad (2.10)$$

This identity shows that the full ambient boundary is not a useful notion for distinguishing exterior boundary conditions: every point of M is a boundary point in $\mathbb{R}^3$.

2.5. Designated Exterior Boundary

Define

$$\partial_o M = M \cap \partial Q_0 = \bigcup_{j=1}^3 \bigcup_{\varepsilon \in \{0,1\}} \Gamma_{j,\varepsilon}, \quad (2.11)$$

where

$$\Gamma_{j,\varepsilon} = M \cap \{x_j = \varepsilon\}. \quad (2.12)$$

Each exterior face is a copy of the standard Sierpiński carpet. At level n , it contains 8^n of the 20^n total cells; hence

$$\mu(\Gamma_{j,\varepsilon}) = \lim_{n \rightarrow \infty} \left(\frac{8}{20}\right)^n = 0, \quad \mu(\partial_o M) = 0. \quad (2.13)$$

Zero measure does not imply zero capacity.

Proposition 2.1 (Capacity of the Exterior Boundary). *Each face $\Gamma_{j,\varepsilon}$ and the exterior boundary $\partial_o M$ have positive 1-capacity:*

$$\text{Cap}_1(\Gamma_{j,\varepsilon}) > 0, \quad \text{Cap}_1(\partial_o M) > 0. \quad (2.14)$$

Proof. Each face contains eight self-similar subcells. The bound (3.12) gives $8\rho_M > 18/5 > 1$, which is the quantitative condition required by the trace theorem for higher-dimensional generalized carpets. Theorems 2.5 and 2.6 of Hino–Kumagai, together with the case described in their Section 5(3), provide a continuous trace map from $\mathcal{F}$ to a Besov space on $\Gamma_{j,\varepsilon}$ and a continuous extension operator [4, Theorems 2.5–2.6 and Section 5(3)]. The normalized self-similar measure on the face is smooth with respect to the associated capacity: it does not charge exceptional sets. If the face had zero capacity, its trace measure would also vanish, contradicting its total mass one. The second inequality follows from $\Gamma_{j,\varepsilon} \subset \partial_o M$ and monotonicity of capacity. $\square$

This positivity makes the exterior Dirichlet condition nontrivial and excludes nonzero constants from the kernel of L_M^D .

2.6. Exterior Strata

The strata relevant to the heat-trace expansion are: the fractal volume M , of dimension $d_0 = d_H$; the exterior faces, of dimension

$$d_1 = \frac{\ln 8}{\ln 3}; \quad (2.15)$$

the edges, of dimension $d_2 = 1$; and the vertices, of dimension $d_3 = 0$.

2.7. Internal Pores and Boundary Choice

The walls of the internal pores are part of the recurrent geometry of the fractal. In the principal realization, they are not interpreted as independent absorbing boundaries. The conservative realization N allows diffusion over all of M ; the exterior Dirichlet realization D imposes killing only on $\partial_o M$. This choice must be distinguished from other possible models with partial absorption or mixed boundary conditions.

3. Canonical Dirichlet Form and Energy Normalization

3.1. Definitions

Let $\mathcal{H}_M = L^2(M, \mu)$. A symmetric Dirichlet form is a pair $(\mathcal{E}, \mathcal{F})$ with $\mathcal{F}$ dense in $\mathcal{H}_M$, and with $\mathcal{E}$ symmetric, nonnegative, closed, and Markovian. The form norm is

$$\|u\|_{\mathcal{E}_1}^2 = \mathcal{E}(u, u) + \|u\|_2^2. \quad (3.1)$$

The form is regular when $\mathcal{F} \cap C(M)$ is dense in $\mathcal{F}$ in the form norm and dense in $C(M)$ in the uniform norm. It is strongly local when $\mathcal{E}(u, v) = 0$ whenever u is constant on a neighborhood of the support of v .

3.2. Canonical Class

The results of Barlow–Bass–Kumagai–Teplyaev and Kajino determine a canonical class of Dirichlet forms on the Menger sponge [2, Theorem 1.2 and Corollary 1.3][5, Theorem 2.4][6, Theorem 4.7].

Theorem 3.1 (Existence and Uniqueness of the Canonical Class). *There exists a regular, symmetric, conservative, and strongly local Dirichlet form $(\mathcal{E}, \mathcal{F})$ on $L^2(M, \mu)$ that is nontrivial and irreducible, and whose class is unique up to multiplication of $\mathcal{E}$ by a positive constant. The form is invariant under global and local cubic symmetries and satisfies the self-similarity relations stated in the next subsection.*

Choose an initial representative and denote it by

$$(\mathcal{E}^{(0)}, \mathcal{F}). \quad (3.2)$$

Its global scale will be fixed in Subsection 3.6. Conservativeness implies

$$1_M \in \mathcal{F}, \quad \mathcal{E}^{(0)}(1_M, v) = 0 \quad (v \in \mathcal{F}). \quad (3.3)$$

Irreducibility, established in [2, Corollary 4.23(a)], gives

$$\ker \mathcal{E}^{(0)} := \{u \in \mathcal{F} : \mathcal{E}^{(0)}(u, u) = 0\} = \text{span}\{1_M\}. \quad (3.4)$$

These properties are independent of the physical interpretation of the model.

3.3. Symmetries and Self-Similarity

Let $\mathcal{G}_0$ be the group of Euclidean isometries preserving Q_0 and M . For $u \in \mathcal{F} \cap C(M)$ and $g \in \mathcal{G}_0$,

$$u \circ g \in \mathcal{F}, \quad \mathcal{E}^{(0)}(u \circ g, u \circ g) = \mathcal{E}^{(0)}(u, u). \quad (3.5)$$

The continuous domain satisfies

$$\mathcal{F} \cap C(M) = \{u \in C(M) : u \circ F_{\mathbf{i}} \in \mathcal{F} \text{ for every } \mathbf{i} \in \mathcal{S}_M\}. \quad (3.6)$$

There exists a unique factor $r_M > 0$, independent of the global scale of the representative, such that

$$\mathcal{E}^{(0)}(u, u) = \frac{1}{r_M} \sum_{\mathbf{i} \in \mathcal{S}_M} \mathcal{E}^{(0)}(u \circ F_{\mathbf{i}}, u \circ F_{\mathbf{i}}), \quad u \in \mathcal{F} \cap C(M). \quad (3.7)$$

The bilinear identity follows by polarization. Equation (3.7) will not be used outside this domain without a density-based extension being justified.

3.4. Renormalization Factors and Dimensions

Define the reciprocal energy factor

$$\rho_M := r_M^{-1}. \quad (3.8)$$

The symbol ρ_M must not be confused with the metric ρ or the effective resistance R_{01} . The self-similar time scale is

$$\tau_M = \frac{20}{r_M} = 20\rho_M. \quad (3.9)$$

The walk dimension is defined by

$$\tau_M = 3^{d_w}, \quad d_w = \frac{\ln(20/r_M)}{\ln 3}. \quad (3.10)$$

With this convention, Definition 2.5 of Kajino identifies $d_w = \log_3(20/r_M)$, and his Theorem 2.7 proves [5, Definition 2.5 and Theorem 2.7]

$$d_w > 2. \quad (3.11)$$

Consequently,

$$\tau_M > 9, \quad r_M < \frac{20}{9}, \quad \rho_M > \frac{9}{20}. \quad (3.12)$$

The spectral dimension is

$$d_s = \frac{2d_H}{d_w} = \frac{2 \ln 20}{\ln(20/r_M)}. \quad (3.13)$$

No exact numerical value is adopted for r_M , d_w , or d_s .

3.5. Effective Resistance Between Opposite Faces

Let

$$A_0 = \Gamma_{1,0}, \quad A_1 = \Gamma_{1,1}. \quad (3.14)$$

For $u \in \mathcal{F}$, let $\tilde{u}$ be a quasi-continuous version, and define

$$\mathcal{A}_{01} = \{u \in \mathcal{F} : \tilde{u} = 0 \text{ q.e. on } A_0, \tilde{u} = 1 \text{ q.e. on } A_1\}. \quad (3.15)$$

Proposition 3.2 (Condenser Between Opposite Faces). *The class $\mathcal{A}_{01}$ is nonempty, closed, and convex in $(\mathcal{F}, \|\cdot\|_{\mathcal{E}^{(0)}})$. Moreover,*

$$0 < \mathcal{C}_{01}(\mathcal{E}^{(0)}) := \inf_{u \in \mathcal{A}_{01}} \mathcal{E}^{(0)}(u, u) < \infty. \quad (3.16)$$

The infimum is attained by a unique energy minimizer h_{01} once the boundary values are fixed quasi-everywhere.

Proof. The faces A_0 and A_1 are compact and disjoint. By Urysohn's lemma, there exists $f \in C(M)$ with $f|_{A_0} = 0$ and $f|_{A_1} = 1$. Regularity provides $v \in \mathcal{F} \cap C(M)$ with $\|v - f\|_\infty < 1/4$. Let $\eta : \mathbb{R} \rightarrow [0, 1]$ be a Lipschitz function such that $\eta = 0$ on $(-\infty, 1/4]$ and $\eta = 1$ on $[3/4, \infty)$. Stability of the domain of a Dirichlet form under Lipschitz compositions implies $\eta \circ v \in \mathcal{F} \cap C(M)$; this function satisfies the two boundary values exactly. Therefore, $\mathcal{A}_{01} \neq \emptyset$.

The subspaces of functions vanishing quasi-everywhere on a closed set are domains of closed part forms [3, Section 4.4]. Hence $\mathcal{A}_{01}$ is a closed and convex affine class. By the Markov property, every minimizing sequence may be truncated to $[0, 1]$ without increasing the energy; it is then bounded in the form norm. Weak compactness in the form Hilbert space, lower semicontinuity, and closedness of the class yield a minimizer.

If the minimum energy were zero, (3.4) would force the minimizer to be constant, contradicting the prescribed values on the positive-capacity sets of Proposition 2.1. If h and g were two minimizers, the parallelogram identity would give $\mathcal{E}^{(0)}(h - g, h - g) = 0$; then $h - g$ would be constant, and its zero boundary values would force $h = g$. This proves positivity and uniqueness. $\square$

Define the effective resistance of the condenser by

$$R_{01}(\mathcal{E}^{(0)}) = \mathcal{C}_{01}(\mathcal{E}^{(0)})^{-1}, \quad (3.17)$$

in accordance with the variational formulation for Dirichlet forms on fractals [12]. If

$$\mathcal{F}_{00} = \{v \in \mathcal{F} : \tilde{v} = 0 \text{ q.e. on } A_0 \cup A_1\}, \quad (3.18)$$

the minimizer satisfies

$$\mathcal{E}^{(0)}(h_{01}, v) = 0, \quad v \in \mathcal{F}_{00}. \quad (3.19)$$

3.6. Normalized Representative and Physical Scale

The canonical class is unique up to scale; the literature does not select a unique dimensional normalization. This manuscript adopts the following reproducible convention based on the opposite-face condenser. The representative is fixed once and for all by

$$\mathcal{E}_M := R_{01}(\mathcal{E}^{(0)}) \mathcal{E}^{(0)}. \quad (3.20)$$

Then

$$R_{01}(\mathcal{E}_M) = 1, \quad \mathcal{C}_{01}(\mathcal{E}_M) = 1. \quad (3.21)$$

From this point onward, $(\mathcal{E}_M, \mathcal{F})$ always denotes the normalized form. This convention fixes the global mathematical energy scale, but not a physical length or mass. The dimensional scale is introduced through

$$\mathcal{L}_M^\# = \Lambda_M^2 L_M^\#. \quad (3.22)$$

Under $\mathcal{E} \mapsto c\mathcal{E}$, the eigenvalues are multiplied by c , whereas r_M , ρ_M , τ_M , d_w , and d_s remain invariant.

4. Self-Adjoint Operators and Boundary Realizations

4.1. Sign Convention and Representation

Adopt

$$L_M^\sharp \geq 0, \quad P_t^\sharp = e^{-tL_M^\sharp}, \quad \Delta_M^\sharp = -L_M^\sharp. \quad (4.1)$$

The representation theorem for closed forms associates with $(\mathcal{E}_M, \mathcal{F})$ a unique nonnegative self-adjoint operator L_M^N such that

$$\mathcal{E}_M(u, v) = \langle L_M^N u, v \rangle_M, \quad (4.2)$$

for $u \in \text{Dom}(L_M^N)$ and $v \in \mathcal{F}$ [3, Theorem 1.3.1 and Corollary 1.3.1].

The operator domain is

$$\text{Dom}(L_M^N) = \{u \in \mathcal{F} : \exists f \in \mathcal{H}_M, \mathcal{E}_M(u, v) = \langle f, v \rangle_M \forall v \in \mathcal{F}\}, \quad (4.3)$$

and $L_M^N u = f$. In addition,

$$\mathcal{F} = \text{Dom}((L_M^N)^{1/2}), \quad \mathcal{E}_M(u, v) = \langle (L_M^N)^{1/2} u, (L_M^N)^{1/2} v \rangle_M. \quad (4.4)$$

4.2. Conservative Realization

Since $\mathcal{E}_M(1_M, v) = 0$,

$$L_M^N 1_M = 0, \quad (4.5)$$

and irreducibility gives

$$\ker L_M^N = \text{span}\{1_M\}. \quad (4.6)$$

The semigroup P_t^N is strongly continuous, self-adjoint, contractive, and Markovian, and satisfies

$$P_t^N 1_M = 1_M. \quad (4.7)$$

The label N is used in the energetic sense and does not presuppose the existence of a classical normal derivative.

4.3. Capacity and Quasi-Continuity

Because $\mu(\partial_o M) = 0$, a boundary condition formulated only μ -almost everywhere would be vacuous. Define

$$\mathcal{E}_{M,1}(u, v) = \mathcal{E}_M(u, v) + \langle u, v \rangle_M. \quad (4.8)$$

The 1-capacity of an open set $O \subset M$ is

$$\text{Cap}_1(O) = \inf\{\mathcal{E}_{M,1}(u, u) : u \in \mathcal{F}, u \geq 1 \text{ } \mu\text{-a.e. on } O\}, \quad (4.9)$$

and it is extended to arbitrary sets by outer regularization. A property holds quasi-everywhere, q.e., if it may fail only on a set of zero capacity. Every function in $\mathcal{F}$ has a quasi-continuous version unique q.e. [3, Theorem 2.1.3].

The result of Section 2 gives

$$\text{Cap}_1(\partial_o M) > 0. \quad (4.10)$$

Thus, prescribing a value q.e. on $\partial_o M$ is a nontrivial energetic condition even though the boundary has zero μ -measure.

4.4. Part Form and Exterior Dirichlet Realization

Let

$$U_M = M \setminus \partial_o M. \quad (4.11)$$

The part form of $(\mathcal{E}_M, \mathcal{F})$ on the relatively open set U_M has domain

$$\mathcal{F}_D = \overline{\mathcal{F} \cap C_c(U_M)}^{\|\cdot\|_{\mathcal{E}_1}}. \quad (4.12)$$

The general theory of part forms gives the exact characterization [3, Theorem 4.4.3]

$$\mathcal{F}_D = \{u \in \mathcal{F} : \tilde{u} = 0 \text{ q.e. on } \partial_o M\}. \quad (4.13)$$

The exterior Dirichlet form is defined by

$$\mathcal{E}_M^D(u, v) = \mathcal{E}_M(u, v), \quad u, v \in \mathcal{F}_D. \quad (4.14)$$

Its associated operator L_M^D is self-adjoint and nonnegative, with

$$\text{Dom}(L_M^D) = \{u \in \mathcal{F}_D : \exists f \in \mathcal{H}_M, \mathcal{E}_M(u, v) = \langle f, v \rangle_M \forall v \in \mathcal{F}_D\}. \quad (4.15)$$

The universal inclusion is $\mathcal{F}_D \subset \mathcal{F}$. In general, one must not claim $\text{Dom}(L_M^D) \subset \text{Dom}(L_M^N)$, because the test spaces are different.

4.5. Kernel, Semigroup, and Domination

If $L_M^D u = 0$, then $\mathcal{E}_M(u, u) = 0$, so u is constant. Since $\tilde{u} = 0$ q.e. on a positive-capacity set, that constant must be zero. Hence

$$\ker L_M^D = \{0\}. \quad (4.16)$$

The semigroup $P_t^D = e^{-tL_M^D}$ is sub-Markovian and corresponds to the part process killed upon leaving U_M ; for continuous paths, exit occurs upon reaching $\partial_o M$. For $f \geq 0$,

$$0 \leq P_t^D f \leq P_t^N f. \quad (4.17)$$

Similarly, for $\alpha > 0$,

$$0 \leq (L_M^D + \alpha I)^{-1} f \leq (L_M^N + \alpha I)^{-1} f. \quad (4.18)$$

The resolvents solve the variational problems

$$\mathcal{E}_M(u_N, v) + \alpha \langle u_N, v \rangle = \langle f, v \rangle_M \quad (v \in \mathcal{F}), \quad (4.19)$$

$$\mathcal{E}_M(u_D, v) + \alpha \langle u_D, v \rangle = \langle f, v \rangle_M \quad (v \in \mathcal{F}_D). \quad (4.20)$$

Self-adjointness alone does not imply spectral discreteness; discreteness will be proved through compactness.

5. Compact Resolvent, Spectral Theorem, and Spectral Positivity

5.1. Heat Kernel and Sub-Gaussian Estimates

The canonical conservative diffusion has a transition density $p_t^N(x, y)$ that is symmetric, nonnegative, and jointly continuous for $t > 0$, with

$$P_t^N f(x) = \int_M p_t^N(x, y) f(y) d\mu(y). \quad (5.1)$$

For small times, it satisfies sub-Gaussian estimates of the form

$$p_t^N(x, y) \asymp \frac{1}{\mu(B(x, t^{1/d_w}))} \exp \left[-c \left(\frac{\rho(x, y)^{d_w}}{t} \right)^{1/(d_w-1)} \right]. \quad (5.2)$$

Ahlfors regularity, $\mu(B(x, r)) \asymp r^{d_H}$, yields

$$p_t^N(x, y) \leq C t^{-d_H/d_w} \exp \left[-c \left(\frac{\rho(x, y)^{d_w}}{t} \right)^{1/(d_w-1)} \right]. \quad (5.3)$$

The measurable Dirichlet kernel satisfies

$$0 \leq p_t^D(x, y) \leq p_t^N(x, y) \quad (5.4)$$

for $\mu \otimes \mu$ -almost every (x, y) .

5.2. Hilbert–Schmidt, Trace Class, and Compactness

For each $t > 0$, the kernel bound and finiteness of $\mu(M)$ imply $p_t^\sharp \in L^2(M \times M)$. Therefore,

$$\|P_t^\sharp\|_{\text{HS}}^2 = \int_{M \times M} |p_t^\sharp(x, y)|^2 d\mu(x) d\mu(y) < \infty, \quad (5.5)$$

and $P_t^\sharp$ is Hilbert–Schmidt and compact. Since

$$P_t^\sharp = P_{t/2}^\sharp P_{t/2}^\sharp, \quad (5.6)$$

it is trace class, with

$$\text{Tr}(P_t^\sharp) = \|P_{t/2}^\sharp\|_{\text{HS}}^2. \quad (5.7)$$

For $\alpha > 0$,

$$(L_M^\sharp + \alpha I)^{-1} = \int_0^\infty e^{-\alpha t} P_t^\sharp dt. \quad (5.8)$$

Compactness is proved in the operator norm. On $(0, \varepsilon)$, the norm of the integral is at most ε because $\|P_t^\sharp\| \leq 1$; on $[\varepsilon, R]$, the integral is a norm limit of sums of compact operators, using norm continuity of the semigroup for $t > 0$; and the tail (R, ∞) has norm at most $e^{-\alpha R}/\alpha$. Letting $\varepsilon \downarrow 0$ and $R \uparrow \infty$ gives

$$(L_M^\sharp + \alpha I)^{-1} \quad \text{compact}. \quad (5.9)$$

This is equivalent to compactness of the embedding of the form domain $\mathcal{F}_\sharp$ into $L^2(M, \mu)$.

5.3. Discrete Spectra

Fix the index sets

$$\mathbb{I}_N = \mathbb{N}_0, \quad \mathbb{I}_D = \mathbb{N}. \quad (5.10)$$

The spectral theorem for compact resolvent yields

$$0 = \lambda_0^N < \lambda_1^N \leq \lambda_2^N \leq \cdots \rightarrow \infty, \quad (5.11)$$

with a simple zero eigenvalue, and

$$0 < \lambda_1^D \leq \lambda_2^D \leq \cdots \rightarrow \infty. \quad (5.12)$$

There are real orthonormal bases $\{\varphi_n^\sharp\}_{n \in \mathbb{I}_\sharp}$ such that

$$L_M^\sharp \varphi_n^\sharp = \lambda_n^\sharp \varphi_n^\sharp. \quad (5.13)$$

For $u = \sum_{n \in \mathbb{I}_\sharp} u_n^\sharp \varphi_n^\sharp$,

$$\|u\|_2^2 = \sum_{n \in \mathbb{I}_\sharp} |u_n^\sharp|^2, \quad (5.14)$$

$$u \in \mathcal{F}_\sharp \iff \sum_{n \in \mathbb{I}_\sharp} \lambda_n^\sharp |u_n^\sharp|^2 < \infty, \quad (5.15)$$

$$\mathcal{E}_M(u, u) = \sum_{n \in \mathbb{I}_\sharp} \lambda_n^\sharp |u_n^\sharp|^2, \quad (5.16)$$

and

$$u \in \text{Dom}(L_M^\sharp) \iff \sum_{n \in \mathbb{I}_\sharp} (\lambda_n^\sharp)^2 |u_n^\sharp|^2 < \infty. \quad (5.17)$$

5.4. Spectral Resolution and Positive Scalar Measure

There exists a projection-valued spectral measure $E^\sharp$ such that

$$L_M^\sharp = \int_{[0, \infty)} \lambda \, dE^\sharp(\lambda). \quad (5.18)$$

For $\psi \in \mathcal{H}_M$, define

$$\mu_\psi^\sharp(B) = \langle \psi, E^\sharp(B) \psi \rangle. \quad (5.19)$$

Since $E^\sharp(B)$ is an orthogonal projection,

$$\mu_\psi^\sharp(B) = \|E^\sharp(B) \psi\|^2 \geq 0. \quad (5.20)$$

In the discrete case,

$$\mu_\psi^\sharp = \sum_{n \in \mathbb{I}_\sharp} |\langle \psi, \varphi_n^\sharp \rangle|^2 \delta_{\lambda_n^\sharp}. \quad (5.21)$$

Positivity concerns the diagonal measures. Mixed measures $\langle \phi, E^\sharp(B) \psi \rangle$ may be complex, and projection kernels need not be positive away from the diagonal.

5.5. Heat Trace, Counting Function, and Min–Max

Define

$$K_M^\sharp(t) = \text{Tr}(e^{-tL_M^\sharp}) = \sum_{n \in \mathbb{I}_\sharp} e^{-t\lambda_n^\sharp}. \quad (5.22)$$

For the Neumann realization,

$$K_M^{N,\perp}(t) = K_M^N(t) - 1 = \sum_{n \geq 1} e^{-t\lambda_n^N}. \quad (5.23)$$

The counting function is

$$N_M^\sharp(\Lambda) = \#\{n \in \mathbb{I}_\sharp : \lambda_n^\sharp \leq \Lambda\}. \quad (5.24)$$

The min–max principle and $\mathcal{F}_D \subset \mathcal{F}$ give

$$\lambda_{k-1}^N \leq \lambda_k^D, \quad k \geq 1. \quad (5.25)$$

The gaps λ_1^N and λ_1^D control exponential semigroup decay on the mean-zero subspace and on the full Dirichlet space, respectively.

6. Heat Trace, Spectral Dimension, and Log-Periodic Structure

6.1. Dimensions and Time Scale

The Hausdorff dimension describes self-similar volume; the walk dimension describes the space–time relation; and the spectral dimension describes thermal and spectral growth:

$$d_H = \frac{\ln 20}{\ln 3}, \quad d_w = \frac{\ln \tau_M}{\ln 3}, \quad d_s = \frac{2d_H}{d_w}. \quad (6.1)$$

Since $d_w > 2$, one has $d_s < d_H$. The discrete time scale is

$$\tau_M = \frac{20}{r_M} = 20\rho_M = 3^{d_w}. \quad (6.2)$$

A spatial contraction by $1/3$ corresponds to a temporal contraction by $1/\tau_M$.

6.2. Leading Heat Growth

The sub-Gaussian estimates imply

$$K_M^\sharp(t) \asymp t^{-d_H/d_w} = t^{-d_s/2}, \quad t \downarrow 0. \quad (6.3)$$

This leading behavior is common to the Neumann and Dirichlet realizations.

6.3. Log-Periodic Expansion

For generalized carpets, Kajino established a finite heat-trace expansion with periodic functions [6, Theorem 4.10 and Remark 4.11]. His formulation may be written with coefficients evaluated at $\ln t$. To work with the positive variable $u = -\ln t$, define $G_{\sharp,k}(u)$ by the reparameterization $u \mapsto -u$; this operation preserves both period and signs. For the three-dimensional sponge, there

are continuous functions $G_{\sharp,k}$, periodic with period $\ln \tau_M$, such that

$$K_M^{\sharp}(t) = \sum_{k=0}^3 t^{-d_k/d_w} G_{\sharp,k}(-\ln t) + O\left(\exp[-c_{\sharp} t^{-1/(d_w-1)}]\right), \quad (6.4)$$

with

$$d_0 = \frac{\ln 20}{\ln 3}, \quad d_1 = \frac{\ln 8}{\ln 3}, \quad d_2 = 1, \quad d_3 = 0. \quad (6.5)$$

Explicitly,

$$\begin{aligned} K_M^{\sharp}(t) = & t^{-d_H/d_w} G_{\sharp,0}(-\ln t) + t^{-d_1/d_w} G_{\sharp,1}(-\ln t) \\ & + t^{-1/d_w} G_{\sharp,2}(-\ln t) + G_{\sharp,3}(-\ln t) + O(e^{-ct^{-1/(d_w-1)}}). \end{aligned} \quad (6.6)$$

The leading term is common and positive:

$$G_{N,0} = G_{D,0} > 0. \quad (6.7)$$

The first boundary correction has opposite signs:

$$G_{N,1} > 0, \quad G_{D,1} < 0. \quad (6.8)$$

No sign is assigned to $G_{\sharp,2}$ or $G_{\sharp,3}$.

6.4. Periodicity, Neumann Reduction, and Nonconstancy

The periodicity relation

$$G_{\sharp,k}(s + \ln \tau_M) = G_{\sharp,k}(s) \quad (6.9)$$

does not imply that the functions are nonconstant. A constant function is also periodic. The proved result is therefore an expansion with periodic coefficients; a nontrivial log-periodic oscillation requires proof that at least one nonconstant coefficient is present.

Define

$$T_M = \ln \tau_M = d_w \ln 3, \quad \omega_M = \frac{2\pi}{\ln \tau_M}. \quad (6.10)$$

The frequency $2\pi/\ln 3$ is not the thermal frequency, because 3 is the spatial scaling factor, whereas τ_M is the temporal scaling factor.

For the reduced Neumann trace,

$$K_M^{N,\perp}(t) = K_M^N(t) - 1. \quad (6.11)$$

This subtraction leaves the terms with $k = 0, 1, 2$ unchanged and replaces only the constant-order coefficient by

$$G_{N,3}^{\perp}(s) = G_{N,3}(s) - 1. \quad (6.12)$$

6.5. Fourier Data and Candidate Complex Exponents

For each periodic coefficient, define

$$\hat{G}_{\sharp,k,m} = \frac{1}{T_M} \int_0^{T_M} G_{\sharp,k}(s) e^{-2\pi i m s/T_M} ds, \quad m \in \mathbb{Z}. \quad (6.13)$$

The associated complex exponents are

$$\beta_{k,m} = \frac{d_k}{d_w} + \frac{2\pi i m}{\ln \tau_M}. \quad (6.14)$$

No pointwise identity between $G_{\sharp,k}$ and its Fourier series is used, because continuity alone does not guarantee pointwise or uniform convergence of that series. The coefficients $\hat{G}_{\sharp,k,m}$ will instead be used through the Mellin transform, whose meromorphic continuation can be obtained directly from periodicity. The actual presence of a nonreal Fourier mode requires

$$\hat{G}_{\sharp,k,m} \neq 0. \quad (6.15)$$

6.6. Counting and Eigenvalue Growth

The two-sided heat-trace bound implies a two-sided bound for the counting function:

$$N_M^\sharp(\Lambda) \asymp \Lambda^{d_s/2}, \quad \Lambda \rightarrow \infty. \quad (6.16)$$

For the upper bound, use $e^{-1}N_M^\sharp(\Lambda) \leq K_M^\sharp(\Lambda^{-1})$. For the lower bound, choose $A > 0$ and split the trace into $\lambda_n \leq A\Lambda$ and $\lambda_n > A\Lambda$; on the tail,

$$e^{-\lambda_n/\Lambda} \leq e^{-A/2} e^{-\lambda_n/(2\Lambda)}, \quad (6.17)$$

so the upper bound for $K_M^\sharp((2\Lambda)^{-1})$ allows A to be chosen such that the tail is smaller than a fixed fraction of the lower bound for $K_M^\sharp(\Lambda^{-1})$. Hence $N_M^\sharp(A\Lambda) \geq c\Lambda^{d_s/2}$.

By monotone inversion,

$$\lambda_n^\sharp \asymp n^{2/d_s} = n^{d_w/d_H}. \quad (6.18)$$

Neither an exact log-periodic law for $N_M^\sharp$ nor a geometric progression of individual eigenvalues follows.

6.7. Internal Ultraviolet Consequence

Since $K_M^\sharp(t) \asymp t^{-d_s/2}$, the internal heat trace diverges as $t \downarrow 0$. The fractal geometry replaces an integer-dimensional power by the spectral dimension and introduces periodic structure in $\ln t$, but it does not by itself produce ultraviolet finiteness.

7. Spectral Zeta Functions and Candidate Poles

7.1. Definitions and Initial Half-Plane

For the Dirichlet realization, define

$$\zeta_M^D(s) = \sum_{n \in \mathbb{L}_D} (\lambda_n^D)^{-s}. \quad (7.1)$$

For the Neumann realization, exclude the zero mode:

$$\zeta_M^{N,\perp}(s) = \sum_{n \geq 1} (\lambda_n^N)^{-s}. \quad (7.2)$$

Both series converge absolutely for

$$\operatorname{Re} s > \frac{d_s}{2}. \quad (7.3)$$

7.2. Mellin Representation

In this half-plane,

$$\zeta_M^D(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} K_M^D(t) dt, \quad (7.4)$$

$$\zeta_M^{N,\perp}(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} K_M^{N,\perp}(t) dt. \quad (7.5)$$

The large-time contribution is entire: $K_M^D(t) = O(e^{-\lambda_1^D t})$ and $K_M^{N,\perp}(t) = O(e^{-\lambda_1^N t})$. Potential singularities arise exclusively from $t \downarrow 0$.

7.3. Meromorphic Continuation: Dirichlet and Reduced Neumann Cases

Proposition 7.1 (Meromorphic Continuation). *The functions ζ_M^D and $\zeta_M^{N,\perp}$ admit meromorphic continuations to all of $\mathbb{C}$. Their possible poles are contained in*

$$\{\beta_{k,m} : k = 0, 1, 2, 3, m \in \mathbb{Z}\}, \quad \beta_{k,m} = \frac{d_k}{d_w} + \frac{2\pi i m}{T_M}. \quad (7.6)$$

Outline of the Proof. The Mellin integral is split into short and long times. The long-time part is entire because of the spectral gap, and the exponentially small remainder in the heat expansion also defines an entire function. For each periodic term, the change of variable $u = -\ln t$ transforms the contribution into a Laplace transform of a periodic function; summation over periods introduces the denominator $1 - e^{-(s-d_k/d_w)T_M}$ and locates the candidates $\beta_{k,m}$. In the Neumann case, subtraction of the constant mode modifies only the stratum term $k = 3$ and restores exponential decay at large times. The complete proof, including local uniform convergence conditions, is given in Appendix B. $\square$

This proof makes the reduced Neumann extension explicit. It is not taken directly from an external theorem but follows from Kajino's expansion and the Mellin argument. In the Dirichlet case, continuation to the full plane and localization of poles are independently established in [7, Theorem 3.1 and Corollary 3.2]; the general result of Steinhurst–Teplyaev uses the convention $\sum \lambda^{-s/2}$, so its pole coordinates must be divided by two when compared with the convention $\sum \lambda^{-s}$ adopted here [8, Theorem 2 and Remark 1].

7.4. Residues and Effective Poles

The residue of $I_{\sharp,k}$ at $\beta_{k,m}$ is

$$\hat{G}_{\sharp,k,m} = \frac{1}{T_M} \int_0^{T_M} G_{\sharp,k}(u) e^{-2\pi i m u/T_M} du. \quad (7.7)$$

Therefore, whenever $\Gamma(\beta_{k,m})$ is finite and the coefficient does not vanish,

$$\operatorname{Res}_{s=\beta_{k,m}} \zeta_M^\sharp(s) = \frac{\hat{G}_{\sharp,k,m}}{\Gamma(\beta_{k,m})}, \quad (7.8)$$

where the reduced zeta function is understood in the Neumann case and, for $k = 3$, the coefficient belongs to $G_{N,3} - 1$.

For $k = 0, m = 0$,

$$\beta_{0,0} = \frac{d_H}{d_w} = \frac{d_s}{2}. \quad (7.9)$$

Since $G_{N,0} = G_{D,0} > 0$ pointwise, its average is positive and the principal pole is effective, with the same positive residue for both realizations.

For $k = 1, m = 0$,

$$s = \frac{d_1}{d_w} \quad (7.10)$$

is an effective pole. Its residue is positive for the Neumann realization and negative for the Dirichlet realization because $G_{N,1} > 0$ and $G_{D,1} < 0$ pointwise.

Nonreal points are effective candidates only when

$$\hat{G}_{\sharp,k,m} \neq 0, \quad m \neq 0. \quad (7.11)$$

Periodicity alone does not guarantee this condition.

7.5. Regularity at Zero and Negative Integer Values

For $k = 3, m = 0$, the candidate is $s = 0$. The singular contribution before multiplication by $1/\Gamma(s)$ is proportional to $1/s$. Since

$$\Gamma(s) = \frac{1}{s} - \gamma_E + O(s), \quad \frac{1}{\Gamma(s)} = s + \gamma_E s^2 + O(s^3), \quad (7.12)$$

one has

$$\frac{1}{s\Gamma(s)} \longrightarrow 1. \quad (7.13)$$

Thus, both unshifted zeta functions are regular at $s = 0$. The remaining candidates $\beta_{k,m}$ have nonnegative real part; in particular, no negative integer is a singularity of the Mellin transform multiplying $1/\Gamma(s)$. Since $1/\Gamma(s)$ vanishes at the nonpositive integers,

$$\zeta_M^D(-q) = \zeta_M^{N,\perp}(-q) = 0, \quad q \in \mathbb{N}. \quad (7.14)$$

The local occurrence of γ_E in this expansion does not by itself yield a universal physical constant.

7.6. Shifted Zeta Function

For $a > 0$, define

$$\zeta_M^\sharp(s; a) = \sum_{n \in \mathbb{I}_\sharp} (\lambda_n^\sharp + a)^{-s} = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} K_M^\sharp(t) dt. \quad (7.15)$$

In the Neumann case, this definition includes the zero mode, regularized by $a > 0$. To justify continuation without interchanging an infinite series with the Mellin integral, fix $Q \in \mathbb{N}_0$ and use the finite expansion

$$e^{-at} = \sum_{q=0}^Q \frac{(-a)^q}{q!} t^q + R_{Q+1}(t), \quad |R_{Q+1}(t)| \leq C_{a,Q} t^{Q+1} \quad (0 < t \leq 1). \quad (7.16)$$

Each finite term shifts the candidates to

$$\beta_{k,m} - q, \quad 0 \leq q \leq Q, \quad (7.17)$$

whereas the remainder is holomorphic in the half-plane $\text{Re } s > d_s/2 - (Q+1)$. Since Q is arbitrary, this procedure meromorphically continues $\zeta_M^\#(s; a)$ to all of $\mathbb{C}$ and locates all candidates at $\beta_{k,m} - q$, $q \in \mathbb{N}_0$. When several contributions do not coincide, the candidate residue is

$$\text{Res}_{s=\beta_{k,m}-q} \zeta_M^\#(s; a) = \frac{(-a)^q}{q!} \frac{\hat{G}_{\#,k,m}}{\Gamma(\beta_{k,m} - q)}. \quad (7.18)$$

If several indices yield the same point, their contributions must be summed; zeros of $1/\Gamma$ may render a candidate removable. One also has

$$\partial_a \zeta_M^\#(s; a) = -s \zeta_M^\#(s+1; a). \quad (7.19)$$

7.7. Determinants and Scale Dependence

Regularity at zero allows the definitions

$$\det_{\zeta} L_M^D = \exp[-(\zeta_M^D)'(0)], \quad \det'_{\zeta} L_M^N = \exp[-(\zeta_M^{N,\perp})'(0)]. \quad (7.20)$$

Under $L \mapsto cL$,

$$\zeta_{cL}(s) = c^{-s} \zeta_L(s), \quad (7.21)$$

$$\zeta'_{cL}(0) = \zeta'_L(0) - (\ln c) \zeta_L(0), \quad (7.22)$$

$$\det_{\zeta}(cL) = c^{\zeta_L(0)} \det_{\zeta} L. \quad (7.23)$$

Thus, $\zeta'(0)$ defines a regularized spectral determinant only after fixing the operator, boundary realization, zero-mode treatment, and scale; it does not by itself define a universal physical constant.

8. Free Scalar Field on $\mathbb{R}^{1,3} \times M$

8.1. Operator-Theoretic Interpretation

The product $\mathbb{R}^{1,3} \times M$ is not an ordinary five-dimensional differentiable manifold. The factor M does not provide a smooth atlas, a classical tangent bundle, or a Dirac operator. The term “five-dimensional” is used here in an operator-theoretic sense: four smooth coordinates plus a fractal internal sector governed by $L_M^\#$.

8.2. Euclidean Formulation

Begin with $X_E = \mathbb{R}^4 \times M$ and

$$\mathcal{H}_E = L^2(\mathbb{R}^4 \times M, dx \, d\mu) \simeq L^2(\mathbb{R}^4) \hat{\otimes} L^2(M, \mu). \quad (8.1)$$

Introduce a physical scale $\Lambda_M > 0$ and a baseline mass parameter $m_0^2 \geq 0$. The physical internal operator is

$$\mathcal{L}_M^\# = \Lambda_M^2 L_M^\#. \quad (8.2)$$

The total quadratic form is

$$\begin{aligned} \mathcal{Q}_E^\sharp[\Phi] &= \int_{\mathbb{R}^4} \int_M \left(|\nabla_x \Phi|^2 + m_0^2 |\Phi|^2 \right) d\mu dx \\ &\quad + \Lambda_M^2 \int_{\mathbb{R}^4} \mathcal{E}_M(\Phi(x, \cdot), \Phi(x, \cdot)) dx. \end{aligned} \quad (8.3)$$

Its natural domain is

$$\mathcal{D}(\mathcal{Q}_E^\sharp) = H^1(\mathbb{R}^4; L^2(M, \mu)) \cap L^2(\mathbb{R}^4; \mathcal{F}_\sharp). \quad (8.4)$$

The form is closed and nonnegative. The associated operator is

$$\mathcal{A}_E^\sharp = -\Delta_{\mathbb{R}^4} \otimes I + I \otimes \Lambda_M^2 L_M^\sharp + m_0^2 I. \quad (8.5)$$

The sum is understood as the self-adjoint sum associated with the closed form; its diagonalization with respect to the internal spectral resolution is a standard case of spectral calculus for strongly commuting self-adjoint operators [13, Chapters VIII and VIII.10].

8.3. Modal Decomposition

Expand

$$\Phi(x, y) = \sum_{n \in \mathbb{I}_\sharp} \phi_n^\sharp(x) \varphi_n^\sharp(y), \quad (8.6)$$

with

$$\phi_n^\sharp(x) = \int_M \Phi(x, y) \varphi_n^\sharp(y) d\mu(y). \quad (8.7)$$

Orthonormality and diagonalization of the energy yield

$$S_E^\sharp[\Phi] = \frac{1}{2} \sum_{n \in \mathbb{I}_\sharp} \int_{\mathbb{R}^4} \left(|\nabla \phi_n^\sharp|^2 + M_{n,\sharp}^2 |\phi_n^\sharp|^2 \right) dx, \quad (8.8)$$

where

$$M_{n,\sharp}^2 = m_0^2 + \Lambda_M^2 \lambda_n^\sharp. \quad (8.9)$$

Equivalently,

$$\mathcal{A}_E^\sharp \simeq \bigoplus_{n \in \mathbb{I}_\sharp} \left(-\Delta_{\mathbb{R}^4} + M_{n,\sharp}^2 \right). \quad (8.10)$$

By the spectral theorem for tensor sums,

$$\sigma(\mathcal{A}_E^\sharp) = \overline{\bigcup_{n \in \mathbb{I}_\sharp} [M_{n,\sharp}^2, \infty)} = [M_{\min,\sharp}^2, \infty), \quad (8.11)$$

where $M_{\min,N}^2 = m_0^2$ and $M_{\min,D}^2 = m_0^2 + \Lambda_M^2 \lambda_1^D$. The numbers $M_{n,\sharp}^2$ are modal thresholds and points at which the multiplicity of the four-dimensional continuum changes; they are not isolated eigenvalues of the product operator. The modal relation is exact within the free model, but it does not identify the modes with known particles.

8.4. Low Modes and Boundary Conditions

For the Neumann realization,

$$M_{0,N}^2 = m_0^2. \quad (8.12)$$

If $m_0 = 0$, the constant mode produces a massless four-dimensional field and requires infrared treatment.

For the Dirichlet realization,

$$M_{1,D}^2 = m_0^2 + \Lambda_M^2 \lambda_1^D > m_0^2. \quad (8.13)$$

Even if $m_0 = 0$, there is a gap $M_{1,D} = \Lambda_M \sqrt{\lambda_1^D}$.

If an eigenvalue has multiplicity g_λ , it generates g_λ fields of equal mass. The multiplicities and representations of the cubic symmetry group must be determined numerically.

8.5. Semigroup, Heat Trace, and Total Spectral Dimension

Strong commutation of the factors gives

$$e^{-t\mathcal{A}_E^\#} = e^{t\Delta_{\mathbb{R}^4}} \otimes e^{-t\Lambda_M^2 L_M^\#} e^{-m_0^2 t}. \quad (8.14)$$

The total kernel factorizes:

$$P_t^{\text{tot},\#}(x, y; x', y') = \frac{e^{-|x-x'|^2/(4t)}}{(4\pi t)^2} p_{\Lambda_M^2 t}^\#(y, y') e^{-m_0^2 t}. \quad (8.15)$$

Because $\mathbb{R}^4$ is noncompact, $e^{-t\mathcal{A}_E^\#}$ does not have a finite total trace. The relevant quantity is the heat-trace density per unit four-dimensional volume:

$$\Theta_M^\#(t) = \frac{e^{-m_0^2 t}}{(4\pi t)^2} K_M^\#(\Lambda_M^2 t). \quad (8.16)$$

Since $K_M^\#(t) \asymp t^{-d_s/2}$,

$$\Theta_M^\#(t) \asymp t^{-(4+d_s)/2}, \quad (8.17)$$

and the spectral dimension defined by the short-time exponent of this density is

$$d_s^{\text{tot}} = 4 + d_s. \quad (8.18)$$

The compact fractal geometry does not automatically improve ultraviolet behavior; it adds spectral density.

8.6. Covariance and Propagators

The bounded inverse exists directly for the Dirichlet realization and, in the Neumann case, when $m_0 > 0$. If $m_0 = 0$ in the Neumann realization, the constant mode requires an infrared prescription or removal before inversion. In the strictly positive cases,

$$C_E^\# = (\mathcal{A}_E^\#)^{-1} = \int_0^\infty e^{-t\mathcal{A}_E^\#} dt, \quad (8.19)$$

with

$$\langle F, C_E^\# F \rangle \geq 0. \quad (8.20)$$

In momentum space, each mode has

$$\tilde{C}_{n,\#}(p) = \frac{1}{p^2 + M_{n,\#}^2}. \quad (8.21)$$

Formally,

$$\tilde{C}_E^\sharp(p; y, y') = \sum_n \frac{\varphi_n^\sharp(y) \varphi_n^\sharp(y')}{p^2 + m_0^2 + \Lambda_M^2 \lambda_n^\sharp}. \quad (8.22)$$

The sum is interpreted operatorially or distributionally after smearing against appropriate test functions.

8.7. Lorentzian Formulation

The formal modal action is

$$S_L^\sharp = \frac{1}{2} \sum_n \int_{\mathbb{R}^{1,3}} \left(\partial_\mu \phi_n^\sharp \partial^\mu \phi_n^\sharp - M_{n,\sharp}^2 (\phi_n^\sharp)^2 \right) d^4x. \quad (8.23)$$

Each mode satisfies

$$(\square_4 + M_{n,\sharp}^2) \phi_n^\sharp = 0, \quad (8.24)$$

and has the Feynman propagator

$$\tilde{G}_{F,n}^\sharp(p) = \frac{i}{p^2 - M_{n,\sharp}^2 + i\varepsilon}. \quad (8.25)$$

8.8. Stability, Reflection Positivity, and Modal Growth

Since $m_0^2 \geq 0$ and $\lambda_n^\sharp \geq 0$, the internal operator generates no tachyonic modes.

Let $\theta(x_0, \mathbf{x}, y) = (-x_0, \mathbf{x}, y)$ be Euclidean time reflection, and let $\mathcal{S}_+$ be a space of test functions supported in $x_0 > 0$ with finite internal expansion.

Proposition 8.1 (Modal Reflection Positivity). *For every finite spectral truncation and every $F \in \mathcal{S}_+$,*

$$\mathfrak{q}_{\text{OS}}^{(N)}[F] := \langle \theta F, C_{E,N}^\sharp F \rangle \geq 0. \quad (8.26)$$

If the covariance of the full tower defines a continuous bilinear form on a completion of $\mathcal{S}_+$ in which finite internal expansions are dense, the inequality extends by continuity to the infinite tower.

Proof. For a finite truncation, $C_{E,N}^\sharp$ is the orthogonal sum of covariances $(-\Delta_{\mathbb{R}^4} + M_{n,\sharp}^2)^{-1}$. The Osterwalder–Schrader form of each free scalar field of nonnegative mass is nonnegative; therefore, their finite sum is also nonnegative. Under the stated continuity hypothesis, a sequence of test functions with finite internal expansion converges to any admissible test function, and the inequality passes to the limit. $\square$

Reflection positivity is only one reconstruction requirement; it does not replace the remaining Osterwalder–Schrader conditions or by itself construct an S -matrix [9, Sections 2–4][10].

The spectral growth yields, as $n \rightarrow \infty$,

$$M_{n,\sharp} \asymp \Lambda_M n^{1/d_s}. \quad (8.27)$$

The factor τ_M characterizes temporal rescaling associated with self-similar cells. It does not imply an exact relation between ordered eigenvalues or a universal ratio between consecutive modal masses.

8.9. Limits of the Scalar Model

The constructed Laplacian acts on functions and does not generate spinors, chirality, gauge charges, or fermionic statistics. The action is quadratic and contains no interactions. A Φ^3 or Φ^4 interaction would require modal tensors built from products of eigenfunctions, which are not computed here. Therefore, the scalar tower is not a leptonic spectrum.

9. Discussion, Physical Scope, and Falsifiability

9.1. Status of the Results

The architecture developed here contains three levels that must not be conflated. The first comprises external results: existence and uniqueness up to scale of the canonical class, energy self-similarity, the inequality $d_w > 2$, sub-Gaussian estimates, and the heat-trace expansion. The second comprises consequences derived in this work: the N/D realizations, compactness of the resolvents, discrete spectra, positive diagonal spectral measures, continuation of the reduced Neumann zeta function, and modal decomposition. The third contains physical hypotheses: the choice of M as an internal sector, the scale Λ_M , and any correspondence between modes and observables.

The validity of the first two levels does not imply the validity of the third. Conversely, a future refutation of the physical interpretation would not invalidate the theorems concerning the canonical Laplacian.

9.2. Dependence on Boundary Conditions, Normalization, and Geometry

The conservative and exterior Dirichlet realizations differ in their zero mode, gap, low eigenvalues, modal masses, and boundary heat terms. The comparison (5.25) shows that they are not two notations for the same model. The boundary condition must be fixed before any phenomenological comparison.

The transformation $\mathcal{E}_M \mapsto c\mathcal{E}_M$ induces $L_M^\sharp \mapsto cL_M^\sharp$ and $\lambda_n^\sharp \mapsto c\lambda_n^\sharp$. The combination entering (8.9) is $\Lambda_M^2 \lambda_n^\sharp$; therefore, neither an isolated eigenvalue nor $\zeta'(0)$ is a universal physical quantity before normalization, boundary realization, zero-mode treatment, and scale are fixed. The choice of the standard sponge is likewise a hypothesis: the article does not establish its physical uniqueness or dynamical selection.

9.3. Scope of the Scalar Model and Ultraviolet Behavior

The internal operator acts on functions. It does not construct spinors, chirality, fermionic statistics, gauge charges, or dynamical gravity. The action considered is quadratic; it contains no vertices, renormalization, scattering amplitudes, or S -matrix. Internal self-adjointness and reflection positivity of finite truncations do not by themselves imply unitarity of a future interacting theory.

The total spectral dimension (8.18) satisfies $d_s^{\text{tot}} > 4$. The fractal sector adds spectral density and does not automatically yield ultraviolet finiteness or exponential suppression of propagators. Likewise, (8.9) defines a scalar tower whose asymptotic growth follows from (6.18); it establishes neither a constant ratio between consecutive masses nor a leptonic identification.

9.4. Claims Excluded from the Present Manuscript

The assignments

$$r_M = \frac{3}{5}, \quad d_s \simeq 1.72, \quad T_M = \ln 3, \quad \lambda_{\text{spec}} = e^{-1}, \quad (9.1)$$

as well as the automatic physical extraction of $2 - \gamma_E$, ultraviolet finiteness, and so-called “exact unitarity,” are explicitly excluded from the set of established results. These expressions could be reintroduced only through independent, reproducible, dimensionally consistent, and noncircular derivations.

9.5. Falsifiability Criteria and the Entry Gate to Paper B

Future evaluation must distinguish the continuous theorem, the discretization, and the physical hypothesis. A numerical discrepancy may reveal an incorrect implementation, lack of convergence, or approximation to a different operator; it does not automatically refute a theorem. Conversely, a convergent construction may be mathematically correct and still lack physical correspondence.

For varying Hilbert spaces, the Kuwae–Shioya framework relates Mosco convergence, resolvents, semigroups, spectral measures, and compact convergence [11, Definition 2.11 and Theorems 2.4, 2.6]. Subsequent work must verify, in this order:

1. explicit definition and normalization of the approximants;
2. variational and resolvent convergence toward L_M^N and L_M^D ;
3. control of eigenvalues, multiplicities, and spectral projections;
4. convergence of heat traces and independent estimation of d_w and d_s ;
5. analysis of possible log-periodic nonconstancy and consistency between heat trace and zeta function;
6. reproducible code, data, uncertainties, and stopping criteria.

Any physical interpretation must be halted if limits depend on subsequences or discretizations, if persistent spurious eigenvalues appear, if signals are smaller than the error, or if normalization introduces the value that was supposed to be discovered. A subsequent phenomenological correspondence will require independently determined parameters, compatible symmetries, and out-of-sample predictions.

10. Conclusions and Subsequent Program

10.1. Structural Synthesis

The construction can be summarized by the chain (1.1). The geometry and measure determine the functional space; the canonical energy class, fixed by (3.20), determines the self-adjoint realizations; compact resolvent yields discrete spectra; and the heat trace and zeta functions organize their scaling structure.

Proposition 10.1 (Main Structural Result). *Let M be the standard Menger sponge with uniform self-similar measure and canonical Dirichlet form normalized by $R_{01} = 1$. For each $\sharp \in \{N, D\}$, the operator $L_M^\sharp$ is self-adjoint, nonnegative, and has compact resolvent. Its semigroup is trace class, and its heat trace admits the expansion (6.4), with continuous periodic coefficients of period $\ln \tau_M$. The functions ζ_M^D and $\zeta_M^{N,\perp}$ admit meromorphic continuations; their poles are contained in the families determined by (6.14), and nonreal poles are effective only when the corresponding Fourier coefficient does not vanish. At the quadratic level, the Euclidean operator (8.5) is unitarily equivalent to a countable tower of free scalar fields with masses given by (8.9).*

10.2. Answer to the Fundamental Question

The requested rigorous architecture consists of a self-similar metric measure space, a normalized canonical Dirichlet form, two boundary realizations, a nonnegative discrete spectrum, a heat-trace expansion with periodic coefficients, meromorphic zeta functions, and a scalar modal decomposition. It can be used as the internal sector of a free model provided that the boundary realization, physical scale, zero-mode treatment, and functional space in which the modal sums converge are specified.

This result establishes a mathematical basis for the TCA-F5D program, but it does not prove that M is a physical dimension, that the modes correspond to particles, or that an interacting extension is finite or unitary. Those questions remain separate from the functional and spectral result.

10.3. Paper B Program

The subsequent work will provisionally be titled:

Numerical Approximation and Spectral Discovery of the Canonical Laplacian on the Menger Sponge

Convergence of Approximants, Spectral Dimension, and Log-Periodic Analysis.

Its objective will be to construct discrete approximants

$$(G_n, \mu_n, \mathcal{E}_n, L_n^N, L_n^D) \tag{10.1}$$

normalized by resistance between opposite faces and to prove that they converge toward the canonical realizations. Eigenvalues, multiplicities, dimensions, and log-periodic coefficients will be computed only after the convergence conditions set out in Section 9 have been met.

The central question of Paper B will be whether an explicit and reproducible sequence of discrete operators can converge toward L_M^N and L_M^D and allow the operator itself to determine its spectrum and diffusion exponents.

10.4. Closing Statement

The result of this article is methodologically prior to any mass theory: it identifies the operator that must be studied before predictions are formulated. The next stage will not consist of fitting a phenomenological formula, but of constructing convergent approximants and subjecting the resulting spectrum to independent mathematical and numerical controls.

The operator determines the spectrum. Discover it; do not impose it.

Prepublication status. This manuscript integrates the transversal mathematical audit, primary bibliographic traceability, scientific editing, internal adversarial review, and final verification of consistency and format. These internal controls do not replace external peer review. This manuscript is the candidate for academic deposit and journal submission.

A. Essential Notation and Conventions

This appendix fixes the minimum dictionary needed to read the article without ambiguity. The expanded symbol dictionary is maintained as an independent supplement to the manuscript.

A.1. Geometry and Measure

Symbol	Definition
$Q_0 = [0, 1]^3$	Initial cube of the self-similar construction.
$\mathcal{S}_M$	Set of the twenty indices of the retained cells.
F_i, F_ω	Elementary similarities and compositions associated with words.
M, M_ω, Q_n	Menger sponge, self-similar cell, and level- n geometric approximant.
ρ	Euclidean metric restricted to M .
μ	Uniform self-similar measure, normalized by $\mu(M) = 1$.
d_H	Hausdorff dimension, $d_H = \ln 20 / \ln 3$.
$\partial_o M$	Designated exterior boundary $M \cap \partial Q_0$.
$\Gamma_{j,\varepsilon}$	Exterior face $M \cap \{x_j = \varepsilon\}$.

A.2. Energy Form and Internal Scales

Symbol	Definition
$(\mathcal{E}^{(0)}, \mathcal{F})$	Initial representative of the canonical class of Dirichlet forms.
$(\mathcal{E}_M, \mathcal{F})$	Representative normalized by $R_{01}(\mathcal{E}_M) = 1$.
$\mathcal{F}_D$	Domain of the part form with quasi-everywhere vanishing on $\partial_o M$.
r_M	Renormalization factor in the self-similar energy identity.
$\rho_M = r_M^{-1}$	Resistance factor associated with the adopted convention.
$\tau_M = 20/r_M$	Self-similar time scale.
$d_w = \ln \tau_M / \ln 3$	Walk dimension.
$d_s = 2d_H/d_w$	Internal spectral dimension.
$R_{01}, \mathcal{C}_{01}$	Effective resistance and conductance between opposite faces.

A.3. Operators and Spectrum

Symbol	Definition
$\sharp \in \{N, D\}$	Conservative or exterior Dirichlet realization.
$L_M^\sharp \geq 0$	Nonnegative self-adjoint operator associated with the corresponding form.
$\Delta_M^\sharp = -L_M^\sharp$	Alternative nonpositive-sign convention for the Laplacian.
$P_t^\sharp = e^{-tL_M^\sharp}$	Heat semigroup.
$p_t^\sharp(x, y)$	Heat kernel, when a kernel version is used.
$\lambda_n^\sharp, \varphi_n^\sharp$	Eigenvalues and orthonormal eigenfunctions.
$E^\sharp(B)$	Projection-valued spectral measure.
$\mu_\psi^\sharp(B)$	Scalar spectral measure $\langle \psi, E^\sharp(B)\psi \rangle$.

Symbol	Definition
$K_M^\sharp(t)$	Heat trace $\text{Tr}(e^{-tL_M^\sharp})$.
$K_M^{N,\perp}(t)$	Reduced Neumann heat trace $K_M^N(t) - 1$.
$N_M^\sharp(\Lambda)$	Spectral counting function.

A.4. Zeta Functions and Scalar Field

Symbol	Definition
$T_M = \ln \tau_M$	Period in the logarithmic time variable.
d_k	Stratum dimensions: $d_0 = d_H$, $d_1 = \ln 8 / \ln 3$, $d_2 = 1$, $d_3 = 0$.
$G_{\sharp,k}$	Periodic coefficients of the heat-trace expansion.
$\widehat{G}_{\sharp,k,m}$	Fourier coefficients of $G_{\sharp,k}$.
$\beta_{k,m}$	Candidate poles $d_k/d_w + 2\pi im/T_M$.
$\zeta_M^D, \zeta_M^{N,\perp}$	Dirichlet zeta function and reduced Neumann zeta function.
Λ_M	External physical scale of the internal operator.
$\mathcal{L}_M^\sharp = \Lambda_M^2 L_M^\sharp$	Internal operator with physical mass-squared dimension.
$\mathcal{A}_E^\sharp$	Total Euclidean operator on $\mathbb{R}^4 \times M$.
$M_{n,\sharp}^2$	Modal mass $m_0^2 + \Lambda_M^2 \lambda_n^\sharp$.

Asymptotic convention. The notation $f(t) \asymp g(t)$ means that positive constants c, C exist such that $c g(t) \leq f(t) \leq C g(t)$ in the stated regime. It does not denote asymptotic equivalence with ratio tending to one.

B. Complete Proof of Meromorphic Continuation

The purpose of this appendix is to justify Proposition 7.1 in detail, particularly the extension of the reduced Neumann zeta function, which is not adopted as a separate external statement.

Proof of Proposition 7.1. Split the Mellin integral over $(0, 1)$ and $(1, \infty)$. The second part is entire because of the spectral gaps. The exponential remainder of the heat expansion also defines an entire function, since

$$\int_0^1 t^{s-1} e^{-ct^{-1/(d_w-1)}} dt \quad (\text{B.1})$$

converges locally uniformly for every $s \in \mathbb{C}$ after a change of variables.

Consider a periodic term with exponent $a_k = d_k/d_w$:

$$I_{\sharp,k}(s) = \int_0^1 t^{s-a_k-1} G_{\sharp,k}(-\ln t) dt. \quad (\text{B.2})$$

With $u = -\ln t$,

$$I_{\sharp,k}(s) = \int_0^\infty e^{-(s-a_k)u} G_{\sharp,k}(u) du. \quad (\text{B.3})$$

Initially for $\text{Re } s > a_k$, periodicity with period T_M allows the intervals $[qT_M, (q+1)T_M]$ to be summed, giving

$$I_{\sharp,k}(s) = \frac{\int_0^{T_M} e^{-(s-a_k)u} G_{\sharp,k}(u) du}{1 - e^{-(s-a_k)T_M}}. \quad (\text{B.4})$$

The numerator is entire, and the denominator vanishes exactly at $s = \beta_{k,m}$. This gives the meromorphic continuation. For the reduced Neumann zeta function, subtraction of the constant mode only replaces $G_{N,3}$ by $G_{N,3} - 1$, and the large-time contribution acquires the gap λ_1^N ; hence the same argument applies. Finally, multiply by $1/\Gamma(s)$. $\square$

Remark on effective poles. The proof locates the set of possible singularities. A point $\beta_{k,m}$ is an effective pole only if the corresponding contribution does not vanish and is not canceled by a zero of $1/\Gamma(s)$. In particular, periodicity of $G_{\sharp,k}$ alone does not imply the existence of nonreal poles.

C. Convergence Protocol for Paper B

This appendix specifies the minimum conditions that a numerical approximation of the canonical operator must satisfy. Its purpose is to prevent an apparent stabilization of eigenvalues from being mistaken for convergence toward $L_M^\sharp$.

C.1. Discrete Objects and Identifications

For each level n , a discrete Hilbert space $\mathcal{H}_n$, a closed nonnegative form $(\mathcal{E}_n^\sharp, \mathcal{F}_n^\sharp)$, its operator $L_n^\sharp$, and identification maps with $L^2(M, \mu)$ must be specified. When the spaces vary, a generalized spectral convergence framework compatible with Kuwae–Shioya will be used [11]. The N and D realizations will be constructed separately and normalized by the same opposite-face resistance criterion.

C.2. Variational Gate

The first requirement is generalized Mosco convergence. One must verify:

1. a lower bound for every sequence converging weakly to the limit;
2. a strongly convergent recovery sequence for every element of the limit domain.

The normalization may not use an assumed value of d_w , d_s , or a spectral ratio as input.

C.3. Operator and Spectral Gate

Variational convergence must imply strong convergence of resolvents and semigroups. To control eigenvalues and multiplicities and prevent spectral pollution, compact convergence, collective compactness, or an equivalent property will also be required. For an isolated spectral interval I , the stable object will be the projection $E_n^\sharp(I)$, not a particular choice of basis within a degenerate eigenspace.

C.4. Heat-Trace Gate

For each $t > 0$, it must be proved or verified with uniform control that

$$\mathrm{Tr}(e^{-tL_n^\sharp}) \longrightarrow K_M^\sharp(t). \quad (\text{C.1})$$

The regime $t \downarrow 0$ requires separation of discretization error, spectral-tail error, and finite-size effects. The estimate of d_s must be cross-checked against $d_s = 2d_H/d_w$, with d_w obtained independently.

C.5. Log-Periodic and Zeta Gate

Nonconstancy of a periodic modulation may be asserted only if its amplitude persists across levels, exceeds the uncertainty, has a period compatible with $T_M = \ln \tau_M$, and yields stable Fourier coefficients. Residues estimated from the heat trace must agree, within controlled errors, with the Mellin transform and truncated zeta functions.

C.6. Reproducibility and Stopping Criteria

The study must publish graphs, weights, measures, matrices, boundary conditions, algorithms, tolerances, software versions, data, and uncertainty analyses. Physical interpretation must be halted if different convergent discretizations produce incompatible limits, if exponents depend on the scheme, if persistent spurious eigenvalues appear, or if the sought signal is smaller than the total error.

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Authorship and Computational Assistance Statement

The physical hypotheses, problem selection, epistemic decisions, interpretation, and scientific responsibility belong to the author. During the reconstruction, artificial-intelligence systems were used as assistants for critique, bibliographic organization, algebraic checking, scientific editing, and LaTeX preparation. Their outputs were treated as drafts subject to verification, not as sources of mathematical authority.

No artificial-intelligence system is listed as an author. External results are cited through primary sources, and original propositions are presented with their hypotheses and proofs. I reviewed Paper A and accept responsibility for its claims, omissions, and possible errors. The Zenodo deposit [14] documents the historical antecedent of the program; the present manuscript explicitly records which claims have been retained, conditioned, or withdrawn.

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